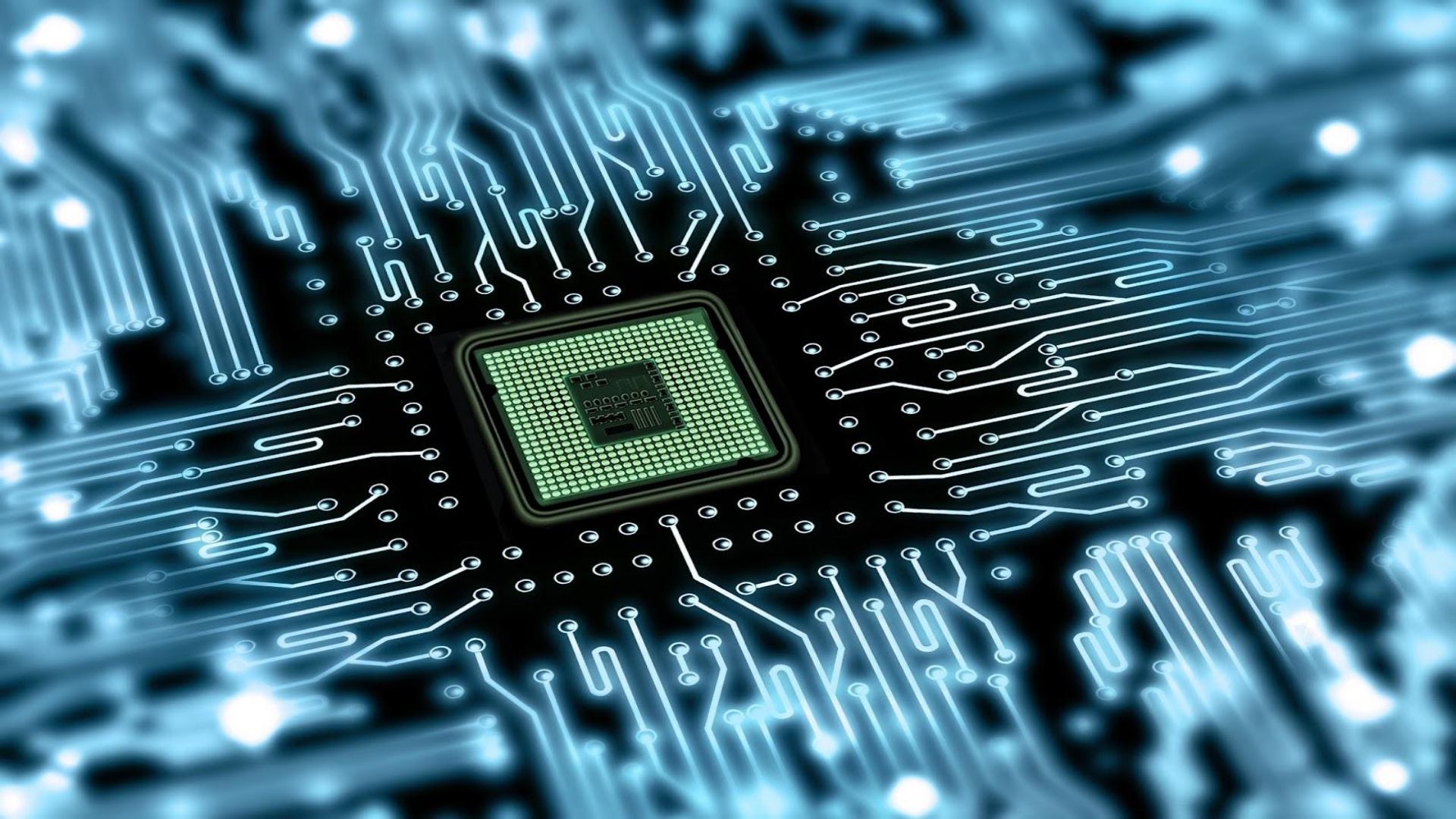


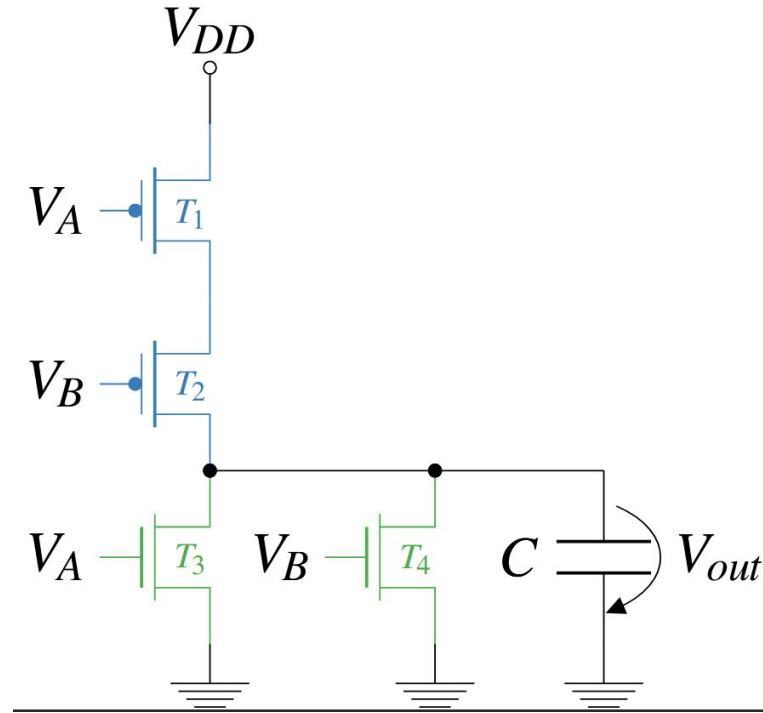
A Symbolic Execution Framework for Symbolic Timing Analysis of Digital Integrated Circuits

Dennis Eigner, Arman Ferdowsi, Ulrich Schmid

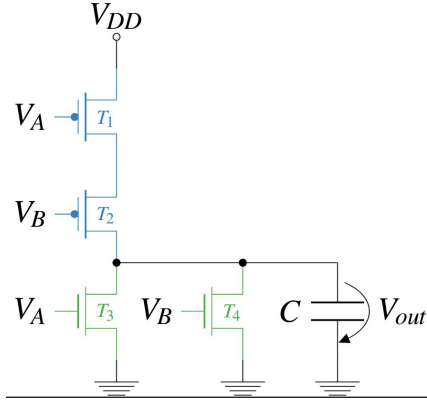
✉ e11808235@student.tuwien.ac.at



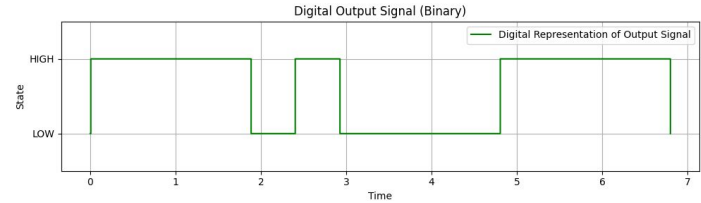
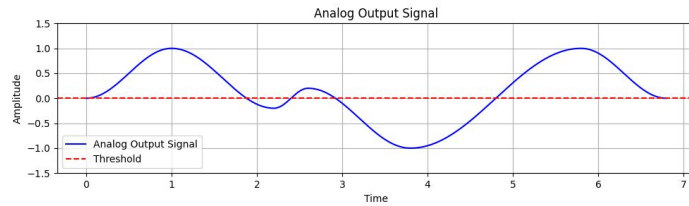
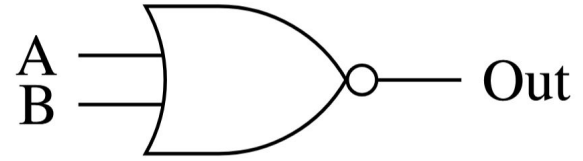
Analog circuits



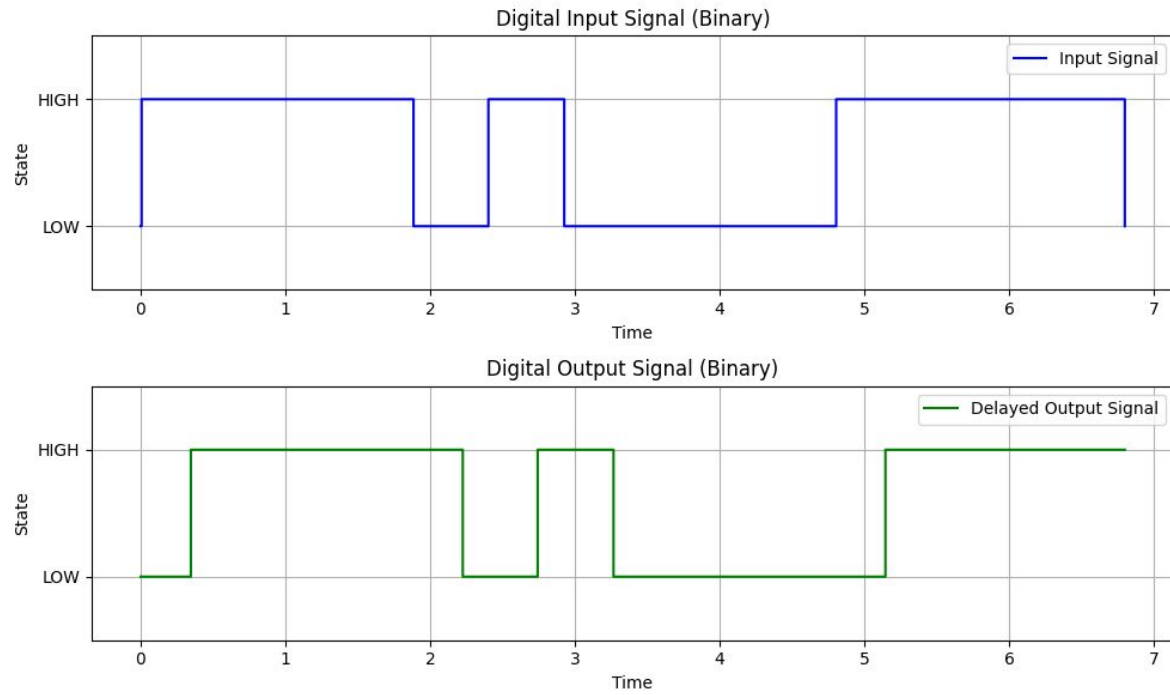
Analog to digital



?



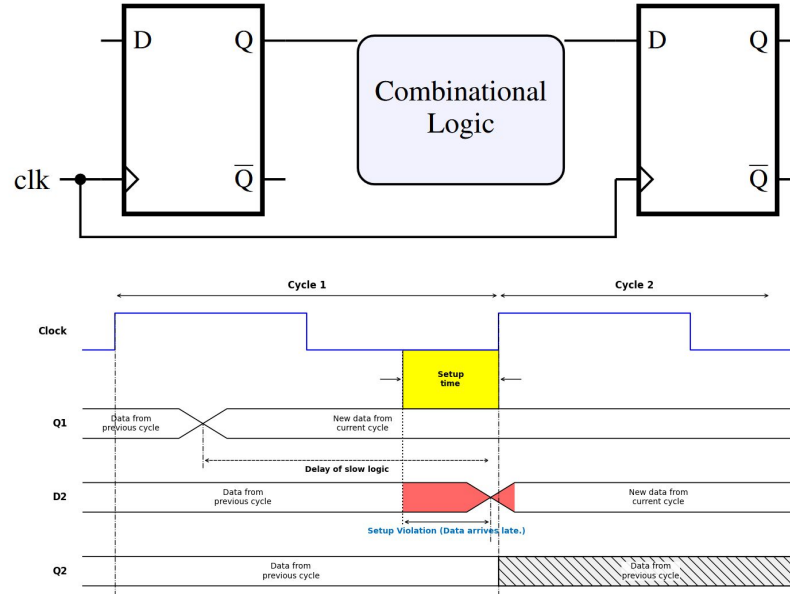
Delays



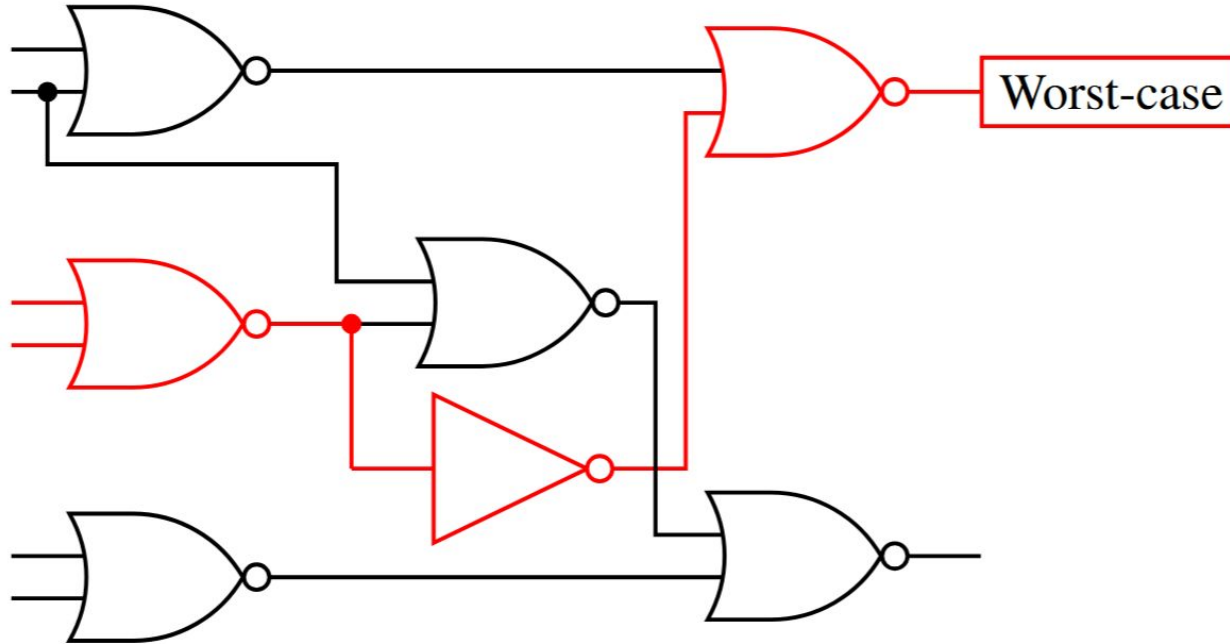
How much delay?

- Really good analog models
- Digital models (pure, inertia) lack accuracy

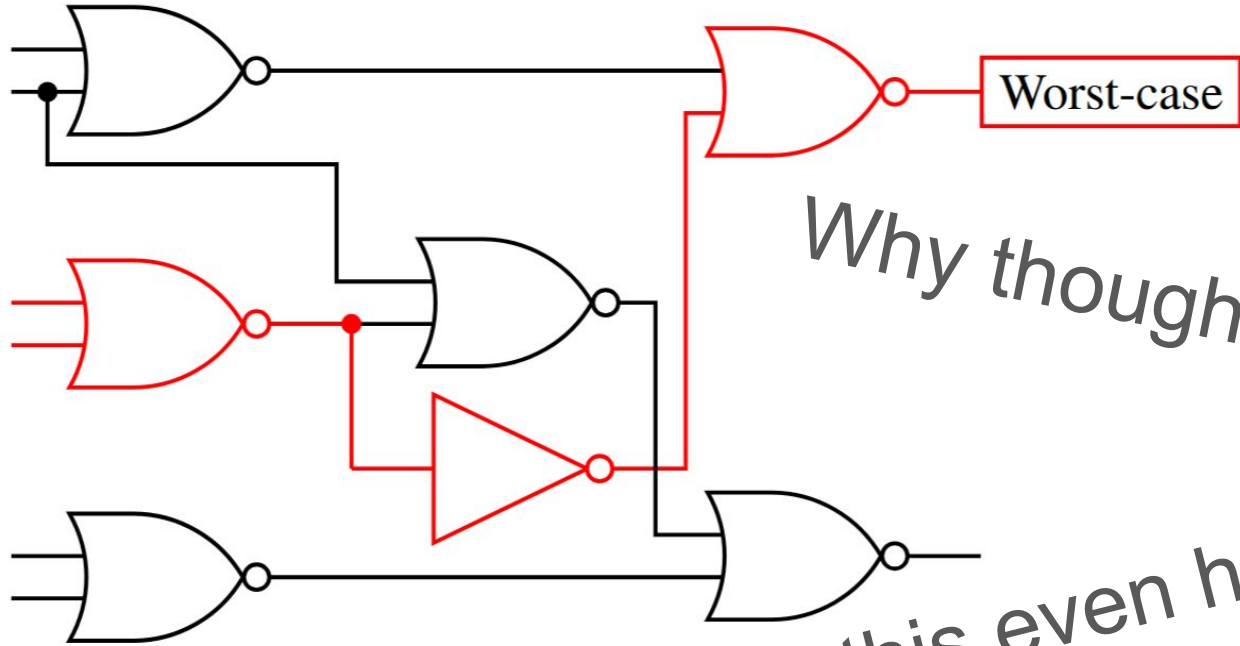
Timing analysis - motivation



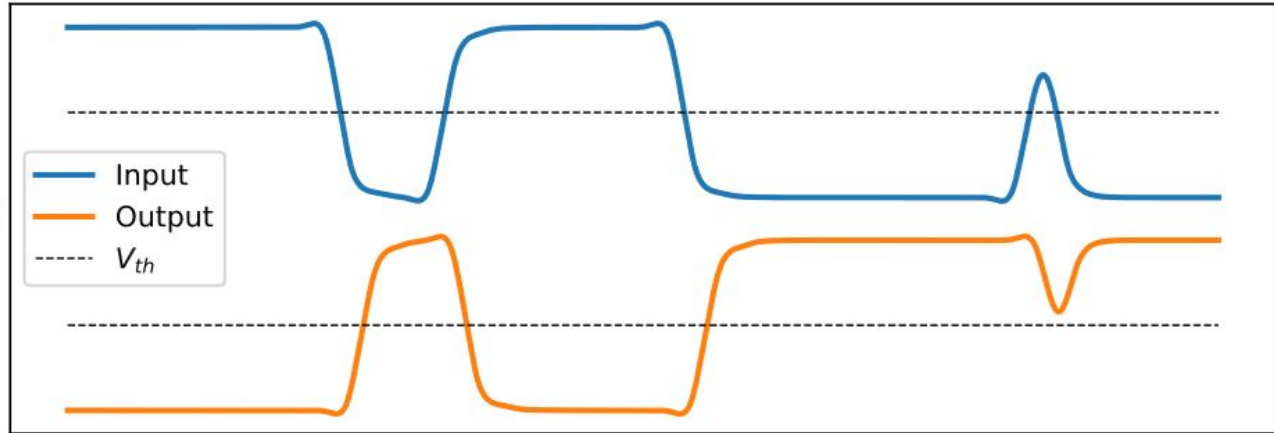
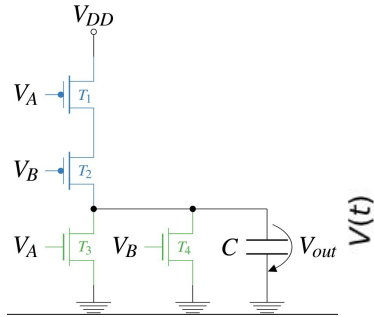
Static Timing Analysis



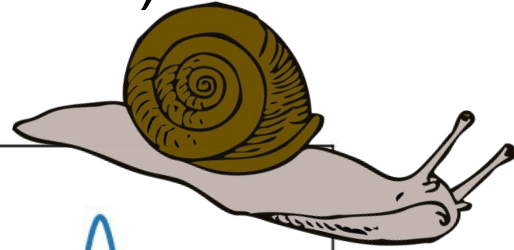
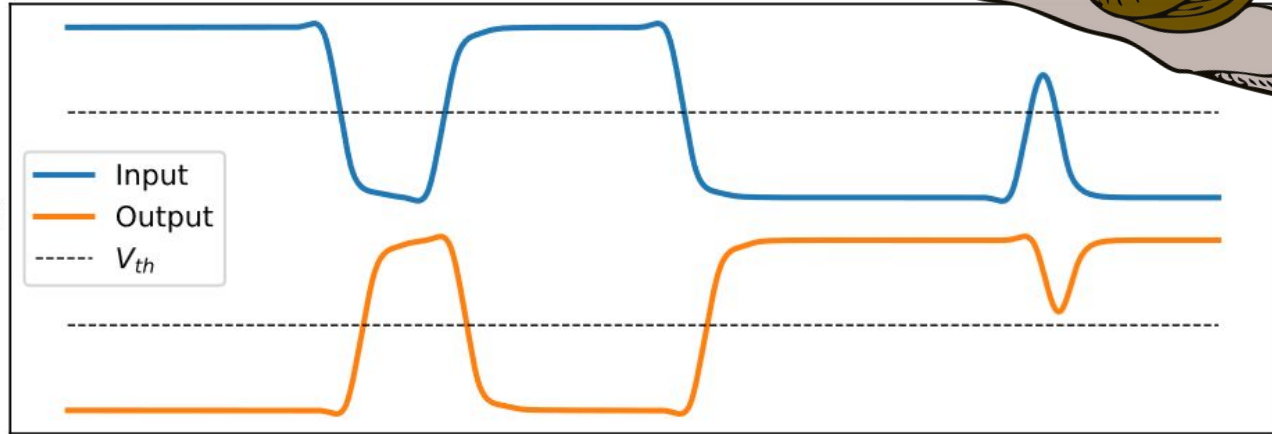
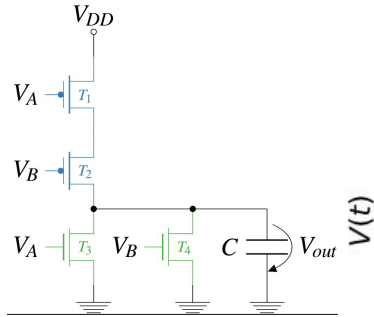
Static Timing Analysis



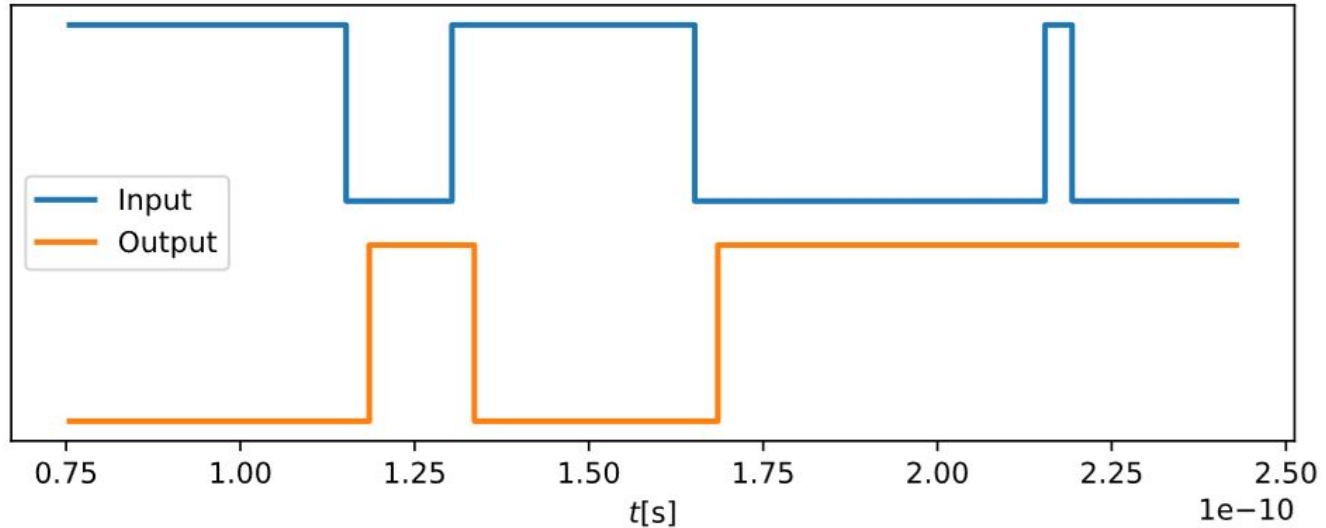
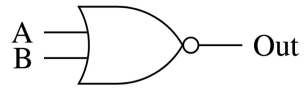
The State Of The Art: Analog Simulation (SPICE)



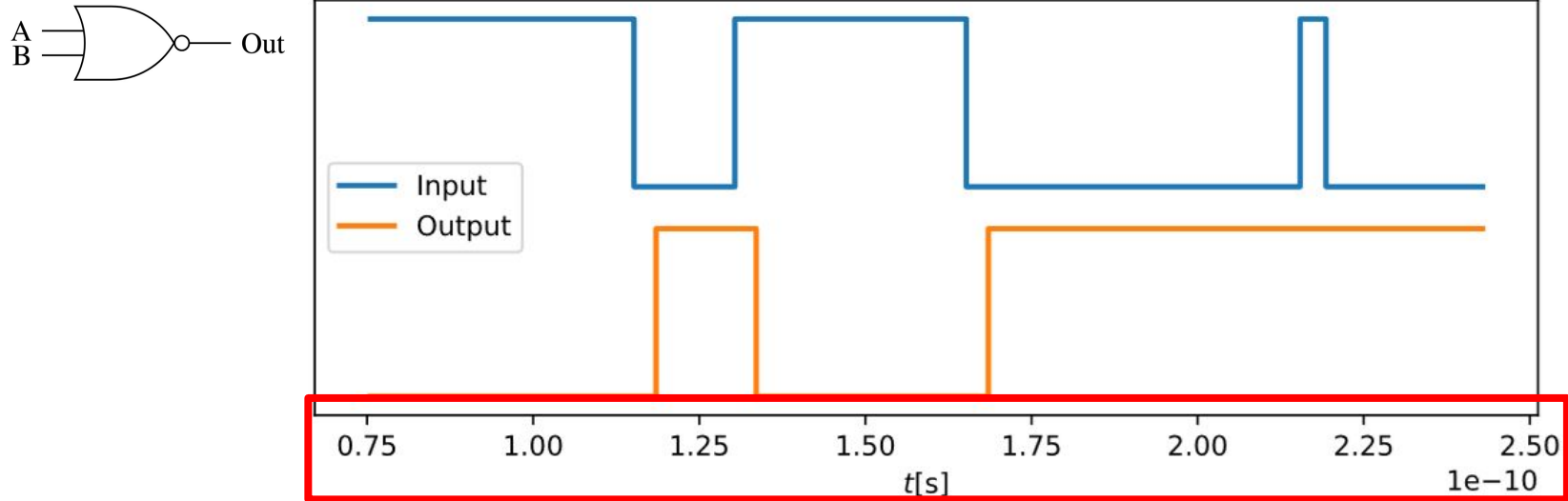
The State Of The Art: Analog Simulation (SPICE)



The State Of The Art: Digital Simulation



The State Of The Art: Digital Simulation



New Delay Models

- Recent advances: history-aware models
- I.e. Ferdowsi et. al [1]
 - Highly accurate digital delay formulas
 - Expressions are **fully symbolic**
 - **Analytic & Closed-form**
- Existing frameworks cannot exploit new formulas

$$\Delta_*(T) = \frac{(\alpha_1 + \alpha_2)(\delta_0(T) - \delta_\infty(T))}{\alpha_1},$$

$$\eta(T) = -2RC_3 \log \left(\frac{1}{2 - \exp\left(-\frac{T + \delta_{\min}}{C'_1 R_{n_B}}\right)} \right) + \delta_{\min}.$$

$$\delta^\uparrow(\Delta, T) \approx \begin{cases} \delta_0(T) - \frac{\alpha_1}{\alpha_1 + \alpha_2} \Delta + \delta_{\min}, & \mathcal{R}_1, \\ \delta_\infty(T) + \delta_{\min}, & \mathcal{R}_2, \\ \eta(T), & \mathcal{R}_3, \end{cases}$$

[1] A. Ferdowsi, U. Schmid, and J. Salzmann, “Drafting and multi-input switching in digital dynamic timing simulation for multi-input gates,” arXiv preprint arXiv:2512.01309, 2025, (This is an extended version of the paper accepted in DATE’26)

Agenda

- **Our Idea:** Formulas instead of numbers
- **Applications:** What can we do with this
- **Our Framework:** The state-space tree
- **Challenge:** Combinatorial explosion, loops
- **Solution:** Symbolic Pruning & meta-transitions

Our Idea: Symbolic Timing Analysis

The Core Idea: Use formulas, not numbers

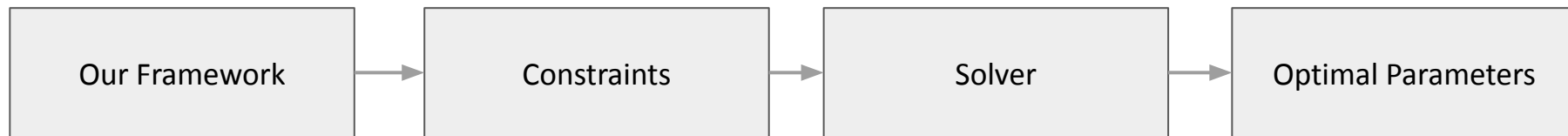
- Express circuit timing behavior in terms of:
 - Symbolic input transition times
 - Symbolic gate parameters
- Compute symbolic delays with a Computer Algebra System

Applications

- Transitions in unwanted order?
- Maximise/minimise an arbitrary goal function

Application 1: Goal-Driven Optimization

- **Goal:** What parameters maximize goal function?
- **Process:**
 - Generate set of candidate paths & symbolic constraints
 - Feed constraints into a solver
 - Solver finds optimal parameters

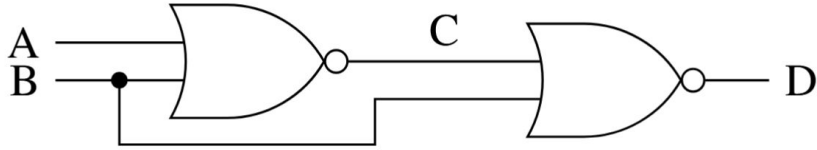


Symbolic Execution Framework

State-Space Tree: Core Concepts

- **Node:** A snapshot of the circuits state
- **Edge:** A single wire transition between states
- **Path:** A sequence of transitions (one possibility)

Building The Tree: Simple Example

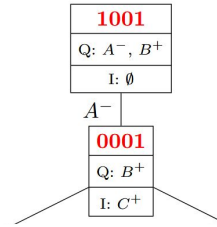
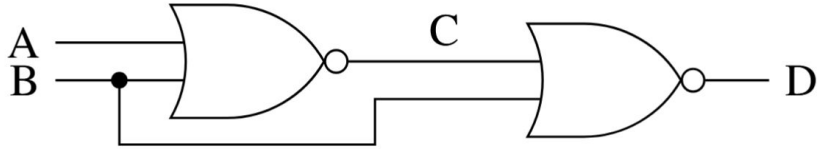


1001
Q: A^-, B^+
I: $\emptyset$

Initial State: $A=1, B=0, C=0, D=1$

Input Queue: A^-, B^+

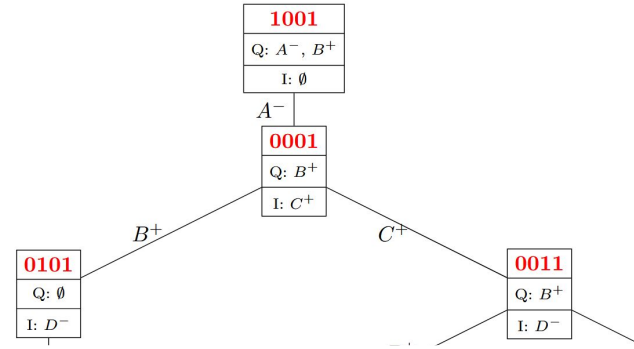
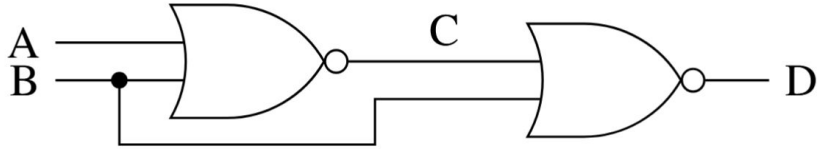
Building The Tree: Simple Example



Initial State: A=1, B=0, C=0, D=1

Input Queue: A⁻, B⁺

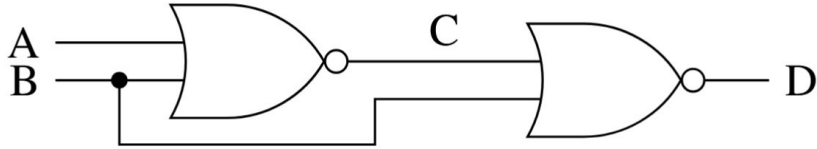
Building The Tree: Simple Example



Initial State: $A=1, B=0, C=0, D=1$

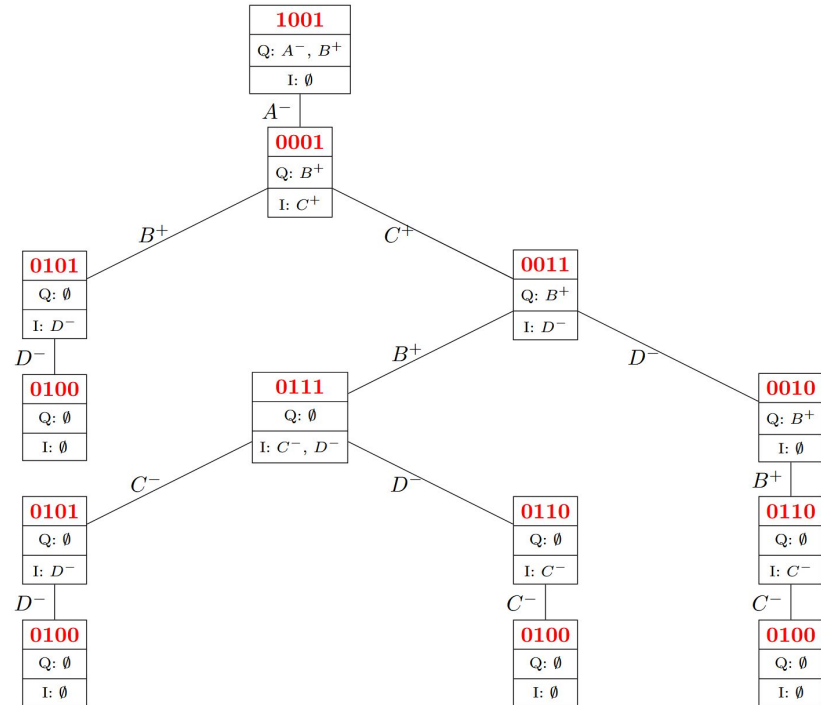
Input Queue: A^-, B^+

Building The Tree: Simple Example

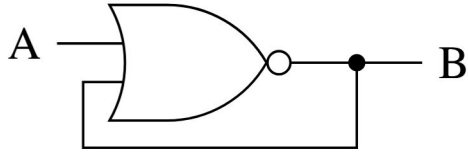


Initial State: A=1, B=0, C=0, D=1

Input Queue: A-, B+



Example 2: A simple loop

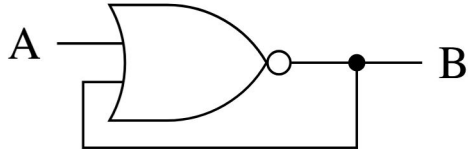


10
Q: A^- , A^+
IQ: $\emptyset$

Initial State: $A=1$, $B=0$

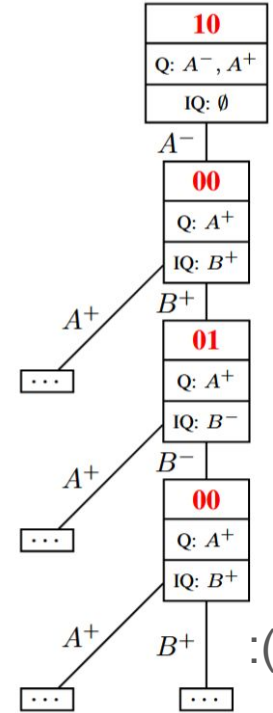
Input Queue: A^- , A^+

Example 2: A simple loop

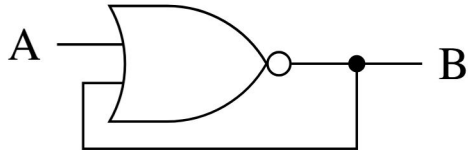


Initial State: A=1, B=0

Input Queue: A-, A+

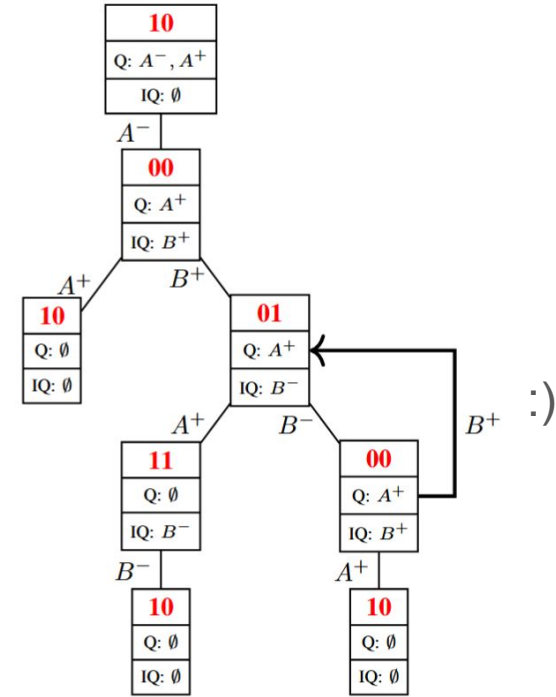


Example 2: A simple loop



Initial State: $A=1, B=0$

Input Queue: A^-, A^+



Challenge: Combinatorial Explosion

- Single transition can trigger many others
 - Precise order is unknown without concrete delay values
- Full exploration has to check **every possible permutation**
- Worst-Case Space complexity:
 - **m**: input queue length
 - **k**: circuit-specific value

$$O\left((k_{max} + 1)^m \cdot k_{max}!\right)$$

Challenge: Combinatorial Explosion

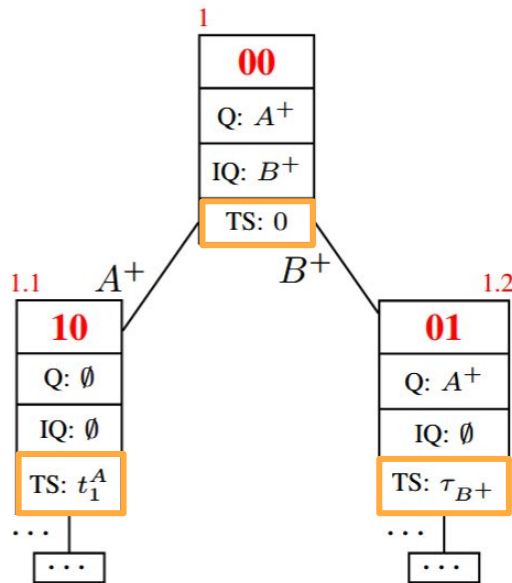
- Single transition can trigger many others
 - Precise order is unknown without concrete delay values
- Full exploration has to check **every possible permutation**
- Worst-Case Space complexity: $O((k_{max})^m)$
 - **m**: input queue length
 - **k**: circuit-specific value
- **Exhaustive exploration is impossible!**

Augmentation to the State Space Tree

- Input transitions happen at $t_1, t_2, \dots$
- Delay model to calculate inconsistency timestamps

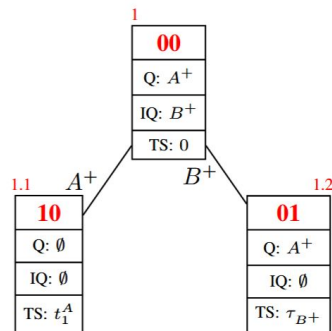
$$\tau_G = P \times H \rightarrow \mathbb{R}$$

- Annotate each node with its timestamp



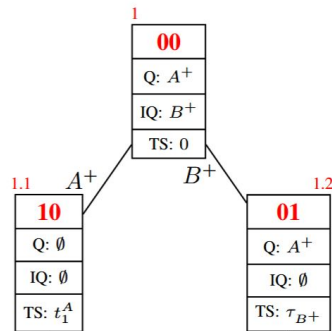
Constraint propagation

- **Idea:** Don't build the whole tree
- Build a set of symbolic constraints



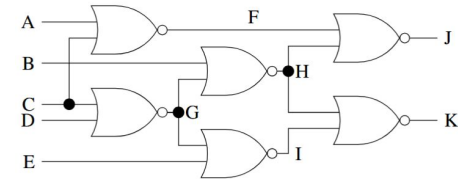
Constraint propagation

- **Idea:** Don't build the whole tree
- Build a set of symbolic constraints



- For path **A+** to occur, delay must satisfy $t_1^A < \tau_{B+}$
- This inequality constraints valid parameter space

Experimental results



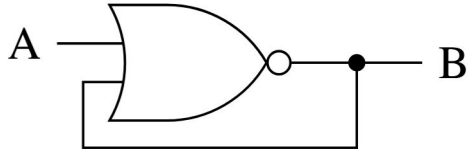
	W.o. Constr. Prop.		With Constr. Prop.		Delta	
 Q 	Et (s)	#C	Et (s)	#C	Et (s)	#C
1	0.095	1	0.141	1	1.48	1
2	0.093	3	0.11	3	1.18	1
4	0.16	112	0.218	15	1.36	0.13
8	185.93	234708	2.801	153	0.015	0.00065
16	/	/	27.283	1991	/	/

|Q| ... Input Queue length

Et (s) ... Execution time (seconds)

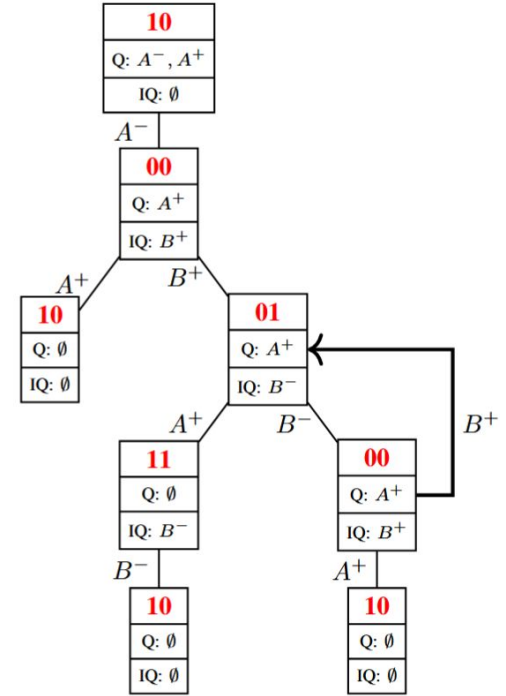
#C ... Number of candidate paths

But loops....

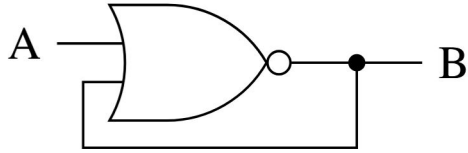


Initial State: $A=1, B=0$

Input Queue: A^-, A^+

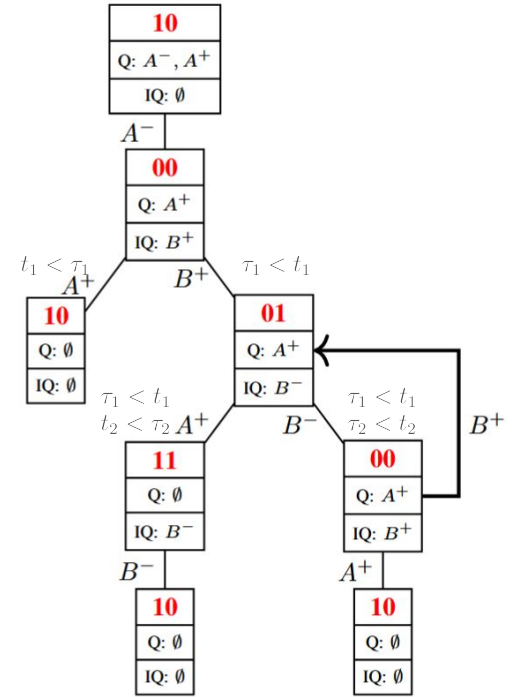


But loops....

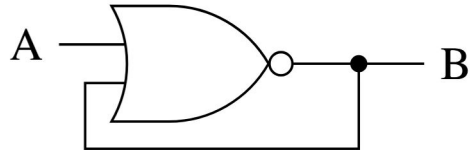


Initial State: A=1, B=0

Input Queue: A-, A+

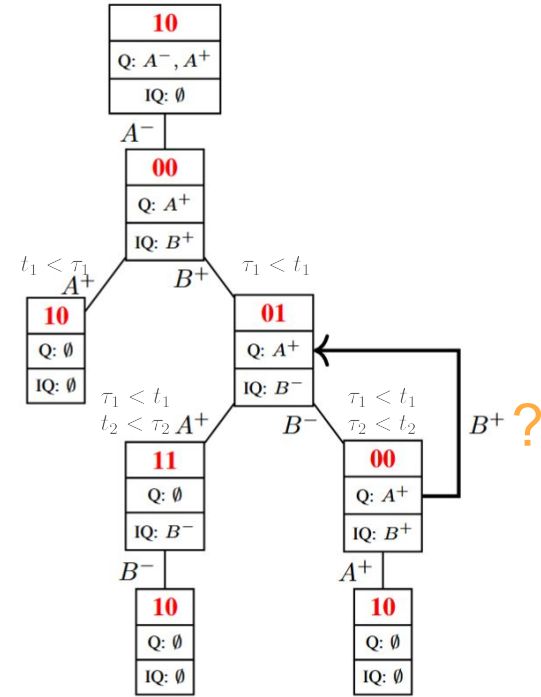


But loops....



Initial State: $A=1, B=0$

Input Queue: A^-, A^+

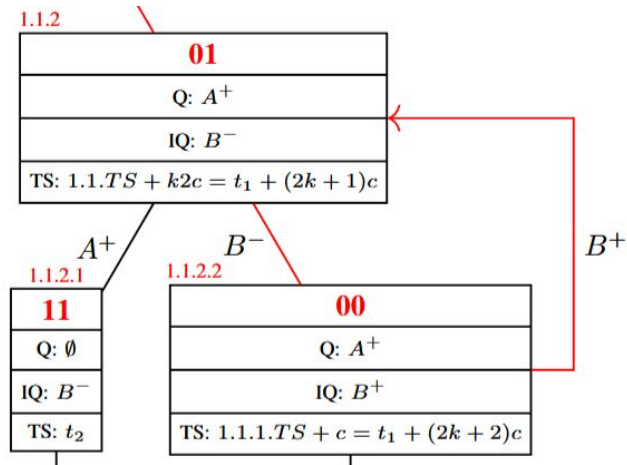


Idea!

- Represent loop delay in closed analytic form
- Replace loop transitions with “meta-transitions”

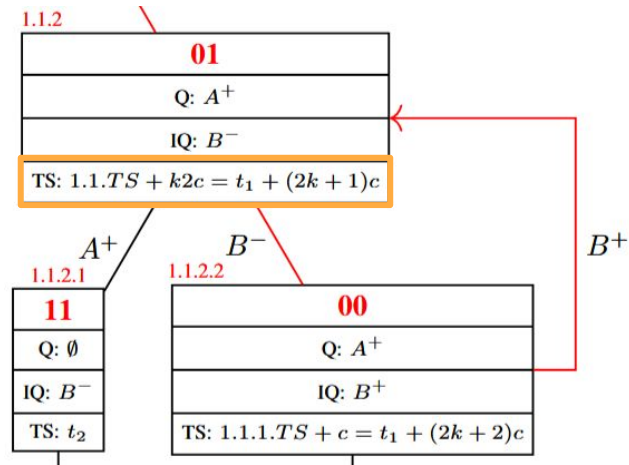
Idea!

- Represent loop delay in closed analytic form
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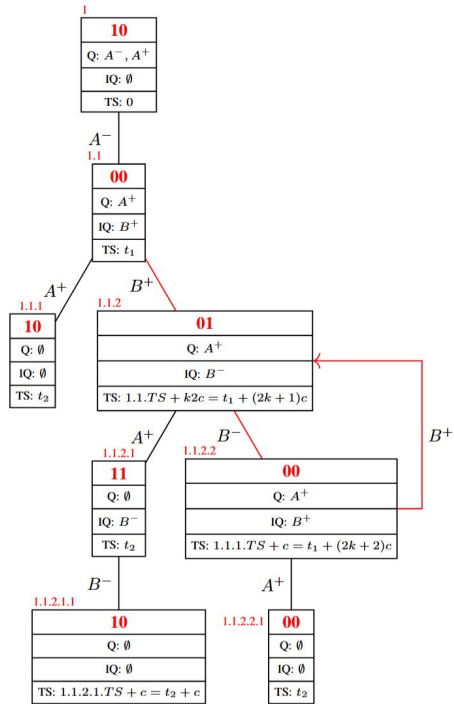


Idea!

- Represent loop delay in closed analytic form
- Replace loop transitions with “meta-transitions”
- Introduce loop counter



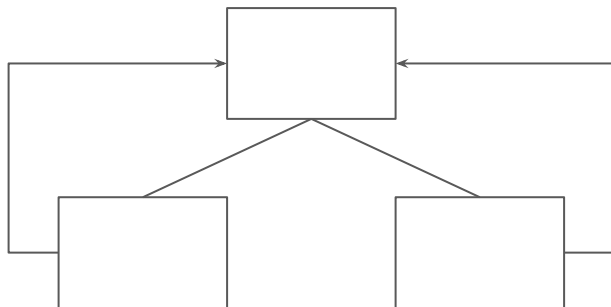
Idea!



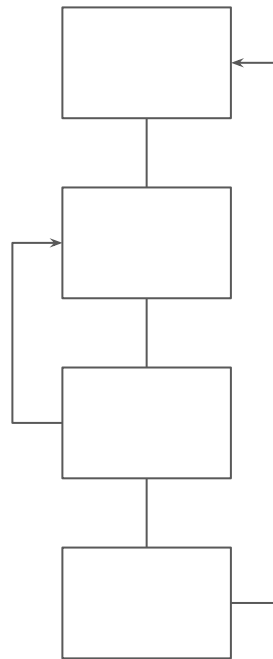
Well...

- Trivial for simple loops

Well...

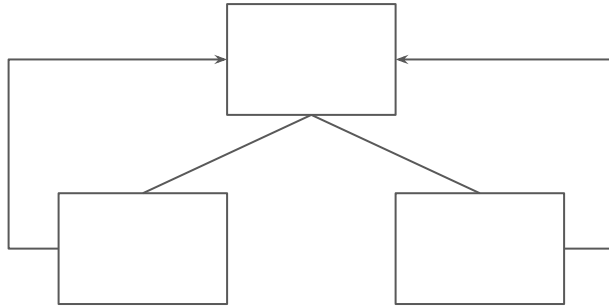


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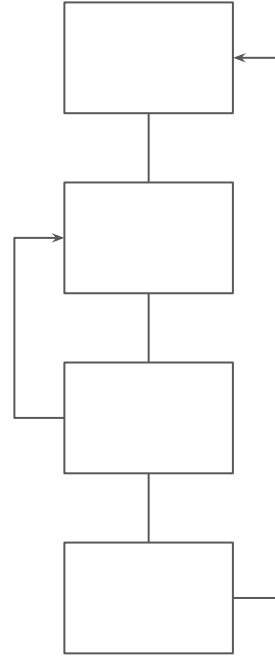


Well...

- Iterative approach doesn't guarantee termination



?



Conclusions & Thank You

Summary

- A symbolic execution framework leveraging new analytic delay models
- Represent the problem as a state-space tree and prune it to reduce complexity
- Represent loops as meta-transitions

Future Work

- Partial order reduction techniques
- Complex loops

Further reading

Eigner, Dennis, Arman Ferdowsi, and Ulrich Schmid. "A Symbolic Execution Framework for Symbolic Timing Analysis of Digital Integrated Circuits." *arXiv preprint arXiv:2608.04036* (2026). <https://arxiv.org/abs/2608.04036>

& Future Work